

SIM & USB Device Management System

Software Requirements Specification (SRS)

Functional & Non-Functional Requirements

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document describes the functional and non-functional requirements for the SIM & USB Device Management System. The system is a web-based application built on ASP.NET Core MVC that enables organizations to track the full lifecycle of SIM cards and USB modems assigned to employees and non-employees, with a complete audit trail of every operation.

1.2 Scope

The system covers the following capabilities:

- Assigning SIM cards and USB modems to employees or non-employees
- Tracking device status changes: active, lost, replaced, returned, or damaged
- Recording device transfers between employees with full history
- Managing Internet quota allocations (base + extra GB)
- Role-based access control via Groups and Permissions
- Complete audit logging of all create / update / delete operations
- Digital or physical receiver signature collection upon device handover

1.3 Stakeholders

Stakeholder	Role	Interest
IT Administrator	System Admin	Full access — manage users, groups, permissions, and devices
HR / Operations Manager	Manager	Assign SIM & USB devices to employees; generate reports
Employee / Non-Employee	End User (recipient)	Receive and sign for devices; view their own subscriptions
Auditor	Read-Only Viewer	Review AuditLog and device history without modifying data
IT Developer	Developer	Build, deploy, and maintain the ASP.NET Core MVC application

1.4 Definitions & Abbreviations

Term	Definition
SIM	Subscriber Identity Module — a card assigned to an employee for mobile internet
USB Modem	A USB device providing mobile broadband internet connectivity
Subscription	A record linking an employee/non-employee to a SIM and/or USB at a point in time
DeviceTransfer	A logged event when a device moves from one employee to another
DeviceStatus	A snapshot of a device's current state (Active / Lost / Replaced / Returned / Damaged)
AuditLog	An immutable record of every data-changing operation, stored as JSON diffs
Quota	The allocated internet data allowance (in GB) for a subscription period
RBAC	Role-Based Access Control — permissions are assigned to Groups, not individuals
EF Core	Entity Framework Core — the ORM used for Code First database access
MVC	Model-View-Controller — the architectural pattern of the web application

2. System Overview

2.1 System Context

The SIM & USB Device Management System is a multi-tier web application. Users interact through a browser-based MVC frontend. The application layer hosts business logic services. Data is persisted in SQL Server via Entity Framework Core Code First migrations. Every data mutation is automatically recorded in the AuditLog table via an EF Core SaveChanges interceptor.

2.2 Technology Stack

Layer	Technology
Frontend	ASP.NET Core MVC · Razor Views · Bootstrap 5 · jQuery
Backend	C# · ASP.NET Core 8 · Dependency Injection · Middleware
ORM / Database	Entity Framework Core 8 (Code First) · SQL Server 2019+
Authentication	ASP.NET Core Identity or custom JWT with RBAC
Audit Logging	EF Core SaveChanges Interceptor → AuditLog table
Architecture	Repository Pattern · Unit of Work · Service Layer · Clean Architecture

3. Functional Requirements

Functional requirements define what the system must do. Each requirement is identified by a unique FR-ID and mapped to its priority and system module.

3.1 Employee & Non-Employee Management

ID	Feature	Description	Priority	Module
FR-01	Add Employee	Admin can create an employee record with Name, Position, and National ID. National ID must be unique.	High	Employee
FR-02	Edit Employee	Admin can update employee details. Changes are logged in AuditLog.	High	Employee
FR-03	Deactivate Employee	Admin can mark an employee as inactive without deleting their history.	Medium	Employee
FR-04	Link to User	Each employee can be linked to one system User account for login access.	High	Employee
FR-05	Add Non-Employee	Admin can register a non-employee (contractor / visitor) with Name, Type, and Contact Info.	Medium	NonEmployee
FR-06	List & Search	Admin can list and search employees/non-employees by name, position, or National ID.	High	Employee

3.2 SIM Card Management

ID	Feature	Description	Priority	Module
FR-07	Register SIM	Admin registers a SIM with Serial Number (ICCID), Phone Number, Network Type, and monthly Fees. Serial number must be unique.	High	SIM
FR-08	Update SIM	Admin can update SIM details. Status changes (Active → Lost / Replaced / Returned) are recorded in DeviceStatus.	High	SIM

ID	Feature	Description	Priority	Module
FR-09	Mark SIM Lost	Admin can mark a SIM as Lost, triggering a DeviceStatus record with date and reporter.	High	SIM
FR-10	Replace SIM	Admin can replace a SIM with a new one. The old SIM's status becomes Replaced and the new SIM ID is referenced.	High	SIM
FR-11	Return SIM	Admin can mark a SIM as Returned from an employee, closing the associated subscription.	Medium	SIM
FR-12	List & Filter	Admin can view all SIMs filtered by Status, Network Type, or assigned employee.	High	SIM

3.3 USB Modem Management

ID	Feature	Description	Priority	Module
FR-13	Register USB	Admin registers a USB modem with Serial Number and Model. Serial number must be unique.	High	USB
FR-14	Update USB	Admin can update USB details and change its status.	High	USB
FR-15	Mark USB Lost	Admin can mark a USB as Lost, creating a DeviceStatus record.	High	USB
FR-16	Replace USB	Admin can replace a USB device, linking old and new serials via DeviceStatus.	High	USB
FR-17	Return USB	Admin can record a USB return from an employee.	Medium	USB
FR-18	List & Filter	Admin can list USB modems filtered by Status or assigned employee.	High	USB

3.4 Subscription Management

ID	Feature	Description	Priority	Module
FR-19	Create Subscription	Admin creates a subscription assigning a SIM (required) and optionally a USB to an employee or non-employee, with a start date and quota.	High	Subscription
FR-20	End Subscription	Admin can end a subscription by setting its EndDate, which triggers device status updates.	High	Subscription
FR-21	View Subscription	Admin and managers can view the full subscription details including linked devices and quota.	High	Subscription
FR-22	Subscription History	The system shows the full subscription history per employee, including past devices and dates.	High	Subscription
FR-23	Record Previous User	When creating a subscription, admin can indicate if the device was previously assigned to another employee.	Medium	Subscription
FR-24	Subscription Notes	Admin can add free-text notes to any subscription for special remarks.	Low	Subscription

3.5 Quota Management

ID	Feature	Description	Priority	Module
FR-25	Define Quota	Admin defines a quota plan with BaseAmount (GB), ExtraAmount (GB), and Period (Monthly/Quarterly).	High	Quota
FR-26	Assign Quota	A quota plan is assigned to a subscription at creation time.	High	Quota
FR-27	Extra Quota	Admin can grant extra GB on top of the base quota for a subscription.	Medium	Quota
FR-28	Quota Period	Quota validity is bounded by ValidFrom and ValidTo dates.	Medium	Quota

3.6 Device Transfer

ID	Feature	Description	Priority	Module
FR-29	Transfer Device	Admin records a device transfer from one employee to another, specifying the SIM and/or USB being moved and the reason.	High	Transfer
FR-30	Auto-Create Sub	On transfer, the system automatically creates a new subscription for the receiving employee and closes the old one.	High	Transfer
FR-31	Transfer History	Admin can view the full transfer history of any SIM or USB device.	High	Transfer
FR-32	Transfer Reason	Admin must specify a reason (Reassigned / Replaced / Lost) when initiating a transfer.	Medium	Transfer

3.7 Receiver Signature

ID	Feature	Description	Priority	Module
FR-33	Capture Signature	Upon device handover, a digital or physical signature is captured and stored against the subscription.	High	Signature
FR-34	Signature Type	System supports two types: Digital (base64 canvas capture) and Physical (scanned image upload).	Medium	Signature
FR-35	Sign Date & User	Signature records the signer's User ID and timestamp automatically.	High	Signature
FR-36	View Signature	Admin can view and download the stored signature for any subscription.	Medium	Signature

3.8 Audit Log

ID	Feature	Description	Priority	Module
FR-37	Auto-Log Changes	Every INSERT, UPDATE, and DELETE on any entity table is automatically captured in AuditLog via an EF Core interceptor.	High	AuditLog
FR-38	Log Old & New	The AuditLog stores both the old JSON snapshot and the new JSON snapshot for every changed record.	High	AuditLog
FR-39	Log Who & When	Each audit entry records the User ID of the performer, the timestamp, and the originating IP address.	High	AuditLog
FR-40	Log Module	The source UI module is recorded in each audit entry for traceability.	Medium	AuditLog
FR-41	Search Audit Log	Admin can filter the audit log by table name, action type, user, or date range.	High	AuditLog
FR-42	Immutable Log	Audit log records cannot be edited or deleted via the application UI.	High	AuditLog

3.9 User Management & Authentication

ID	Feature	Description	Priority	Module
FR-43	Create User	Admin creates system users with Username, Email, and a hashed password.	High	Auth
FR-44	Login	Users authenticate with Username and Password. Failed attempts are logged.	High	Auth
FR-45	Password Hash	Passwords are never stored in plain text — bcrypt hashing is applied.	High	Auth
FR-46	Assign to Group	Each user belongs to exactly one Group which defines their permissions.	High	Auth
FR-47	Last Login	The system records the last successful login timestamp per user.	Low	Auth

ID	Feature	Description	Priority	Module
FR-48	Deactivate User	Admin can deactivate a user account without deleting their audit history.	High	Auth

3.10 Role-Based Access Control (RBAC)

ID	Feature	Description	Priority	Module
FR-49	Create Group	Admin creates permission groups (e.g. IT Admin, Manager, Viewer).	High	RBAC
FR-50	Define Permissions	Permissions are defined by Controller + ActionName pairs (e.g. SubscriptionController.Create).	High	RBAC
FR-51	Assign Permissions	Admin assigns one or more permissions to a group via GroupPermission.	High	RBAC
FR-52	Enforce Permissions	The system enforces permissions on every controller action — unauthorized access returns 403.	High	RBAC
FR-53	Permission Check	Permission checks are performed server-side, not only in the UI.	High	RBAC

3.11 Reporting & Export

ID	Feature	Description	Priority	Module
FR-54	Device Report	Admin can generate a report of all devices with their current status and assigned employee.	High	Reports
FR-55	Employee Report	Admin can view a report showing all devices ever assigned to a specific employee.	High	Reports
FR-56	Quota Report	Admin can view quota usage across all active subscriptions.	Medium	Reports
FR-57	Export to Excel	All report tables can be exported to Excel (.xlsx) for offline use.	Medium	Reports

4. Use Case Summary

UC-ID	Use Case	Actor	Description
UC-01	Assign SIM + USB to Employee	IT Administrator	Admin selects an employee, chooses available SIM and USB, sets quota and start date, then saves — subscription is created and audit logged
UC-02	Mark Device as Lost	IT Administrator	Admin selects the device, sets status to Lost, adds notes and reporter — DeviceStatus record is created and subscription may be closed
UC-03	Transfer Device to New Employee	IT Administrator	Admin selects source subscription, chooses target employee — old subscription is ended, new subscription is opened, DeviceTransfer is logged
UC-04	Replace SIM with New One	IT Administrator	Admin marks old SIM as Replaced, registers or selects new SIM, links them in DeviceStatus, updates subscription
UC-05	Capture Receiver Signature	IT Administrator	After device handover, admin opens subscription, captures digital signature on canvas or uploads scanned copy — ReceiverSignature saved
UC-06	View Employee Device History	Manager / Admin	User searches for an employee — system shows all past and current subscriptions, devices, transfers, and status events
UC-07	Review Audit Log	Auditor / Admin	Auditor filters the audit log by date, table, or user — views JSON diff of old vs new values for each operation
UC-08	Manage Users & Permissions	IT Administrator	Admin creates users, assigns them to groups, configures which permissions each group holds
UC-09	Generate Device Status Report	Manager / Admin	Admin opens reports module — views all devices with status,

UC-ID	Use Case	Actor	Description
			assigned employee, subscription dates — exports to Excel
UC-10	Login to System	Any User	User enters credentials — system validates, records LastLogin, redirects to role-appropriate dashboard

5. Non-Functional Requirements

Non-functional requirements define the quality attributes and constraints the system must satisfy regardless of specific behaviour.

5.1 Performance

ID	Category	Requirement	Metric / Target	Priority
NFR-01	Performance	All standard page loads must complete within the defined threshold under normal load	< 2 seconds (p95)	High
NFR-02	Performance	Database queries on indexed columns (NationalID, SerialNumber) must respond within threshold	< 200 ms	High
NFR-03	Performance	Report generation for up to 10,000 subscription records must complete within threshold	< 5 seconds	Medium
NFR-04	Performance	The system must support concurrent users without degradation	≥ 50 concurrent users	High
NFR-05	Performance	AuditLog writes must not block the primary transaction — use async interceptor	Non-blocking	High

5.2 Security

ID	Category	Requirement	Metric / Target	Priority
NFR-06	Security	Passwords must be hashed using a strong algorithm — never stored in plain text	bcrypt (cost ≥ 12)	High
NFR-07	Security	All pages require authenticated sessions; unauthenticated requests redirect to login	401 → redirect	High
NFR-08	Security	All controller actions must enforce server-side RBAC permission checks	403 on violation	High

ID	Category	Requirement	Metric / Target	Priority
NFR-09	Security	All user inputs must be validated and sanitized to prevent SQL Injection and XSS	OWASP Top 10	High
NFR-10	Security	Session tokens must expire after a defined period of inactivity	30 minutes idle	High
NFR-11	Security	HTTPS must be enforced for all traffic in production	TLS 1.2+	High
NFR-12	Security	Audit log records must be immutable — no UI delete or update is permitted	Append-only	High
NFR-13	Security	Failed login attempts must be limited to prevent brute-force attacks	5 attempts → lockout	Medium

5.3 Reliability & Availability

ID	Category	Requirement	Metric / Target	Priority
NFR-14	Reliability	The system must maintain high uptime during business hours	≥ 99.5% uptime	High
NFR-15	Reliability	All database operations must be wrapped in transactions — partial saves are not permitted	ACID compliance	High
NFR-16	Reliability	Device transfer must atomically close old subscription and open new one in a single transaction	0 orphaned records	High
NFR-17	Availability	Scheduled maintenance windows must be communicated and occur outside business hours	Off-peak only	Medium
NFR-18	Recoverability	The system must support database backups with a defined recovery point objective	RPO ≤ 24 hours	High

5.4 Maintainability

ID	Category	Requirement	Metric / Target	Priority
NFR-19	Maintainability	All C# classes must follow Clean Architecture — presentation, service, and data layers are separated	Layer isolation	High
NFR-20	Maintainability	All entity models must use EF Core Code First — schema changes managed via migrations	EF Core Migrations	High
NFR-21	Maintainability	Repository and Unit of Work patterns must be used for all data access	No direct DbContext in controllers	High
NFR-22	Maintainability	Global exception handling middleware must catch and log all unhandled exceptions	No raw 500 pages	High
NFR-23	Maintainability	Service layer must use Dependency Injection registered in Program.cs	DI container	High
NFR-24	Maintainability	Code must include XML documentation comments on all public methods and classes	Standard XML docs	Medium

5.5 Usability

ID	Category	Requirement	Metric / Target	Priority
NFR-25	Usability	The UI must be responsive and usable on desktop, tablet, and mobile screens	Bootstrap 5 responsive	High
NFR-26	Usability	All forms must display inline validation messages on submission errors	Client + server validation	High
NFR-27	Usability	All data tables must support pagination — no unbounded result sets displayed	Max 25 rows/page	Medium
NFR-28	Usability	Search and filter controls must be present on all list views	All list pages	High

ID	Category	Requirement	Metric / Target	Priority
NFR-29	Usability	Success and error feedback must be shown via toast notifications after every operation	Toast UI feedback	Medium
NFR-30	Usability	Sensitive actions (delete, deactivate) must require a confirmation dialog before executing	Confirmation modal	High

5.6 Scalability

ID	Category	Requirement	Metric / Target	Priority
NFR-31	Scalability	The database schema must support at least 100,000 subscription records without schema changes	100K+ records	Medium
NFR-32	Scalability	All foreign key columns used in joins must have database indexes	Indexed FKs	High
NFR-33	Scalability	The AuditLog table must be designed for append-only high-volume writes	Partitionable by date	Medium
NFR-34	Scalability	The application must be deployable as a containerized service for horizontal scaling	Docker-ready	Low

5.7 Compliance & Data Integrity

ID	Category	Requirement	Metric / Target	Priority
NFR-35	Compliance	All personally identifiable information (NationalID, Email) must be protected at rest and in transit	Encryption required	High
NFR-36	Compliance	National ID must be validated for format before storage	Format validation	High

ID	Category	Requirement	Metric / Target	Priority
NFR-37	Data Integrity	Deleting a SIM or USB that has active subscriptions must be prevented at the application and DB level	Referential integrity	High
NFR-38	Data Integrity	All decimal fields (Fees, Quota amounts) must use decimal(10,2) to prevent floating-point errors	decimal(10,2)	High
NFR-39	Data Integrity	All datetime fields must store UTC time and display in the user's local timezone	UTC storage	Medium

6. Constraints & Assumptions

6.1 Technical Constraints

- Backend must be implemented in C# using ASP.NET Core MVC 8
- Database must be SQL Server 2019 or later
- ORM must be Entity Framework Core 8 with Code First approach
- No third-party paid services — only open-source or built-in ASP.NET Core features
- Authentication must not rely on external OAuth providers (standalone login only)

6.2 Business Assumptions

- An employee can only have one active subscription per device type (one active SIM, one active USB) at a time
- Non-employees can receive devices temporarily — their subscriptions must have an EndDate
- Device Serial Numbers are treated as permanent identifiers — they cannot be changed after registration
- The AuditLog is write-only from the application perspective — no business process requires deleting log entries
- All monetary amounts (Fees) are in a single currency — no multi-currency support is required

7. Revision History

Version	Date	Changes	Author
1.0.0	28/06/2026	Initial SRS — full functional and non-functional requirements	Mohamed Hassan El-Dokany